MODULE II

2	**Solar Energy:** Fundamentals; Solar Radiation; Estimation of solar radiation on horizontal and inclined surfaces; Solar radiation Measurements - Pyrheliometers, Pyranometer, Sunshine Recorder. Solar Thermal systems: concentrating and non-concentrating collectors - Flat plate collectors; Solar tower electric power plant. Photovoltaic system for electric power generation – Classification of PV system - Principle of Solar cell, advantages, disadvantages and applications of solar photovoltaic system.	9

WHY SOLAR ENERGY

- Non-conventional source of energy
- Readily available
- Free
- Non-polluting
- Solar energy received in the form of radiation, can be converted directly or indirectly into other forms of energy, such as heat and electricity.
- Energy supplied by the sun in one hour is almost equal to the amount energy required by the human population in one year.

- Draw backs of application of solar energy
- 1) intermittent and variable manner in which it arrives at the earth’s surface
- 2) large area required to collect the energy at a useful rate

WHAT IS SOLAR ENERGY

- Originates with the **thermonuclear reactions** occurring in the sun.
- Energy is radiated by the sun as the electromagnetic waves of which 99 percent have wavelengths in the range of 0.2 to 4.0 micrometers.
- Solar energy reaching top of earth's atmosphere consist of

  8% UV radiation (short wave length , <0.39 micrometer)

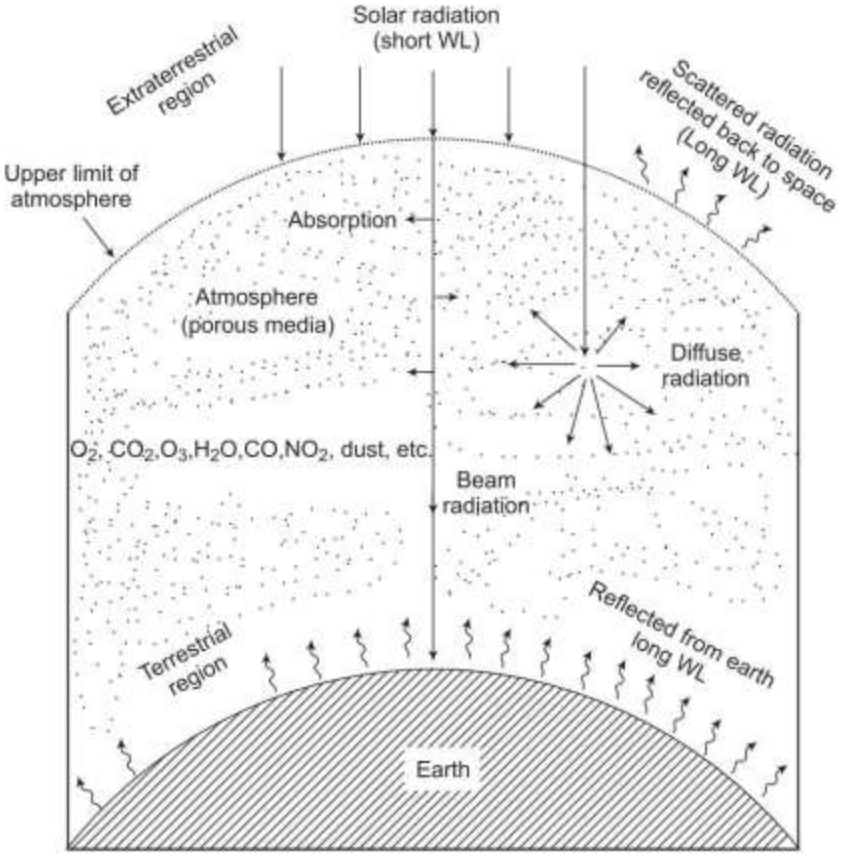
  46% visible light (0.39-0.78 micro meter)

  46% infrared radiation (long wave length , >0.78 micrometer)

SOLAR CONSTANT

- The rate at which Solar energy arrives at the top of the atmosphere is called the Solar Constant (I_{sc}).
- It is defined as the energy received from the sun per unit time on a unit surface area perpendicular to the sun's direction at the mean distance of earth from the sun
- The standard value National aeronautics and space administrations (NASA)-1353 watts/sq.metre.

Extra-terrestrial and Terrestrial Radiations



Extra-terrestrial and Terrestrial Radiations

- The intensity of solar radiation keeps on attenuating with distance from the sun
- The distance between sun and earth varies due to the elliptical motion of the earth

- Extra-terrestrial Radiation (I_{ext}): Solar radiation incident on the outer atmosphere

$$I_{ext} = I_{sc}\left[1 + 0.033\cos\left(\frac{360}{365}n\right)\right] \quad \text{W/m}^2$$

Here, n is the day of the year counted from 1st January.

- Terrestrial Radiation/Solar Irradiation: Solar radiation that reaches Earth
- Solar insolation: Solar radiation energy received per unit area (kWh/m^2)

- The extra-terrestrial(outside the earth's atmosphere) radiation can be determined by using solar constant as follows:

$$I_{ext} = I_{sc} * (R_{av}/R) \text{ W/m}$$

where

I_{ext} is the extra-terrestrial radiation

I_{sc} is the solar constant

R_{av} is the mean distance between the sun and the earth

R is the actual sun-earth distance

SOLAR RADIATION

- In earth atmosphere, part of radiation is reflected back in to the space especially by cloud.
- Radiation entering the atmosphere is partly absorbed by molecules in the air, also some part is scattered by droplets in clouds and by dust particles
- Solar radiation that has not been absorbed or scattered and reaches directly from the sun is called direct radiation

SOLAR RADIATION

- Solar Radiation = Beam Radiation + Diffuse Radiation
- **Beam Radiation (Direct radiation)** – Reaches directly to the earth surfaces without change in direction
- **Diffused Radiation** – Solar radiation from the sun after its direction has been changed by reflection and scattering by atmosphere
- Total solar radiation received at any point on earth surface = Beam Radiation + Diffuse Radiation

- Also called Insolation: Total solar radiation received on a horizontal surface of unit area(1 sq.m) on the ground in unit time (1 day).
- Insolation varies with date, time, altitude of sun in the sky and with weather conditions (clouds, humidity).

- Sun at Zenith: It is the position of sun directly overhead
- Irradiance(W/m2): The rate of incident energy per unit area of a surface
- Albedo: The earth reflects back 30% of the total solar radiant energy to the space by reflection from clouds, by scattering and by reflection at the earths surface. This is called the albedo of the earths atmosphere system.

ABSORPTION AND SCATTERING

- *Variation in solar radiation reaching the earth than received at the outside of the atmosphere is due to Absorption and Scattering*
- *Absorption* : Short wave Ultra-violet rays by ozone and long wave infra red by CO_2 and moisture in the atmosphere.
- *Scattering* : As Solar radiation passes through the earth's atmosphere the components of the atmosphere, such as water vapour, dust in the atmosphere causes scattering.

BASIC SUN EARTH ANGLES

Latitude or angle of latitude(λ)

- The latitude of a location on earth's surface is the angle made by the radial line joining the specified location to the centre of earth with the projection of this line on the equatorial plane.
- The latitude at equator is zero and it is 90^0 at poles.

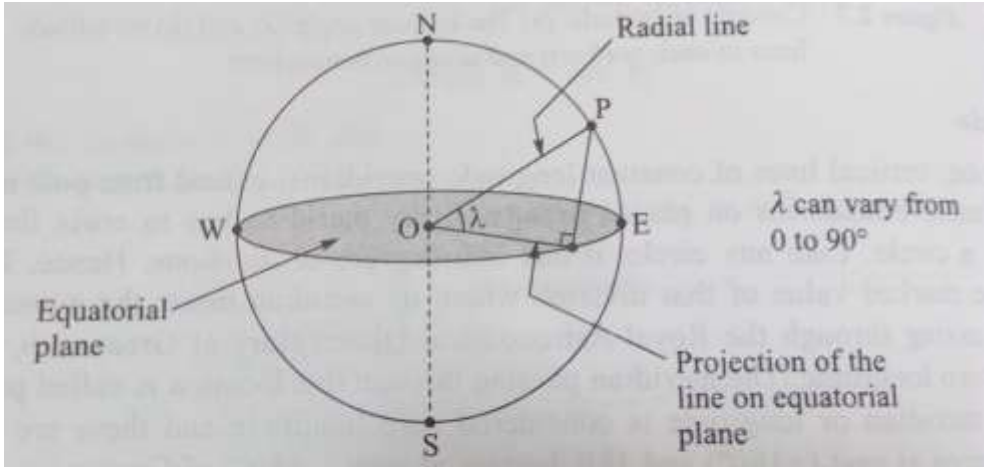


Figure 2.9 The angle of latitude

Declination angle(δ)

- It is the angle made by the line joining the centres of sun and earth with the equatorial plane.

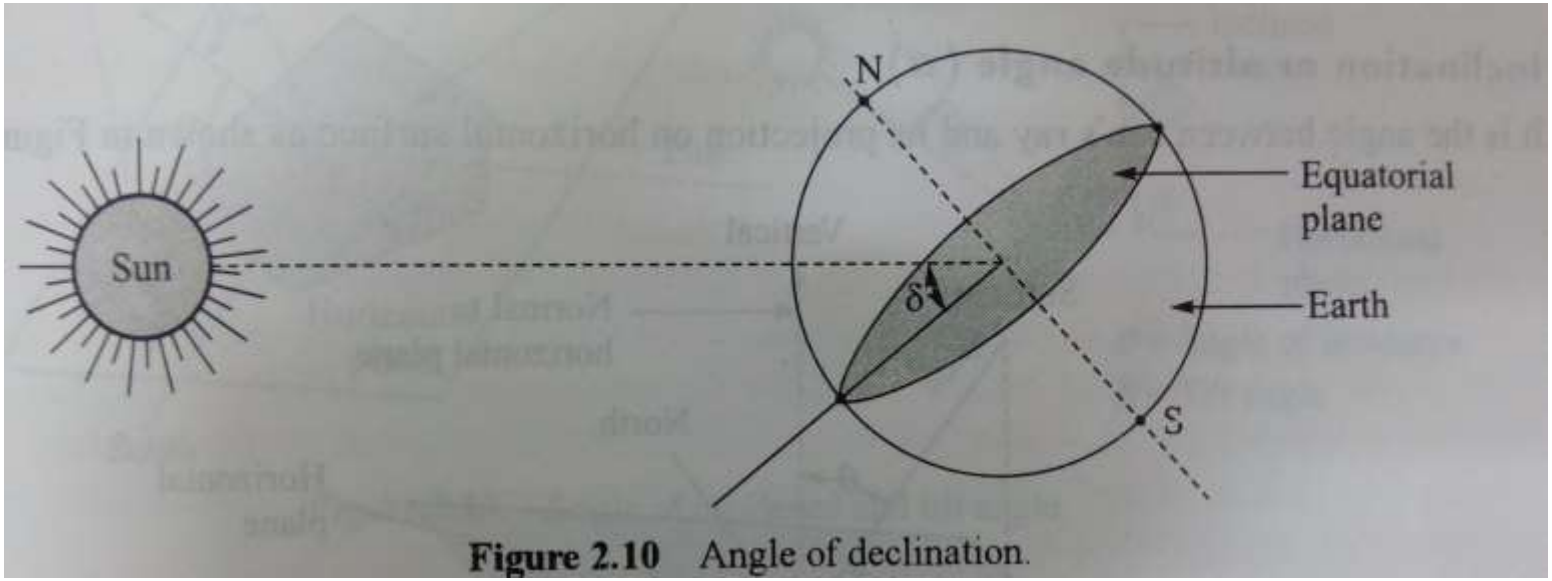


Figure 2.10 Angle of declination.

- The angle of declination varies when the earth revolves around the sun.
- For minimum or maximum declination the sun appears to stand still

- It has maximum value of 23.45º when earth achieves a position in its orbit corresponding to 21 June and it has minimum value of -23.45º when earth is in orbital position corresponding to 22 December.
- The angle of declination is taken positive when it is measured above the equatorial plane in the northern hemisphere.
- The angle of declination can be given by

$$\delta = 23.45 \times \sin\left[\frac{360}{365}(284 + n)\right]$$

where n is the number of days counted from 1 January

Hour angle(ω)

- It is an angular measure of time at the rate of 15 degree per hour
- The hour angle at any instant is the angle through which the earth has to turn to bring the meridian of a point directly in line with sun's rays.
- Hour angle is measured from Noon , based on local apparent time being positive in afternoon and negative in forenoon

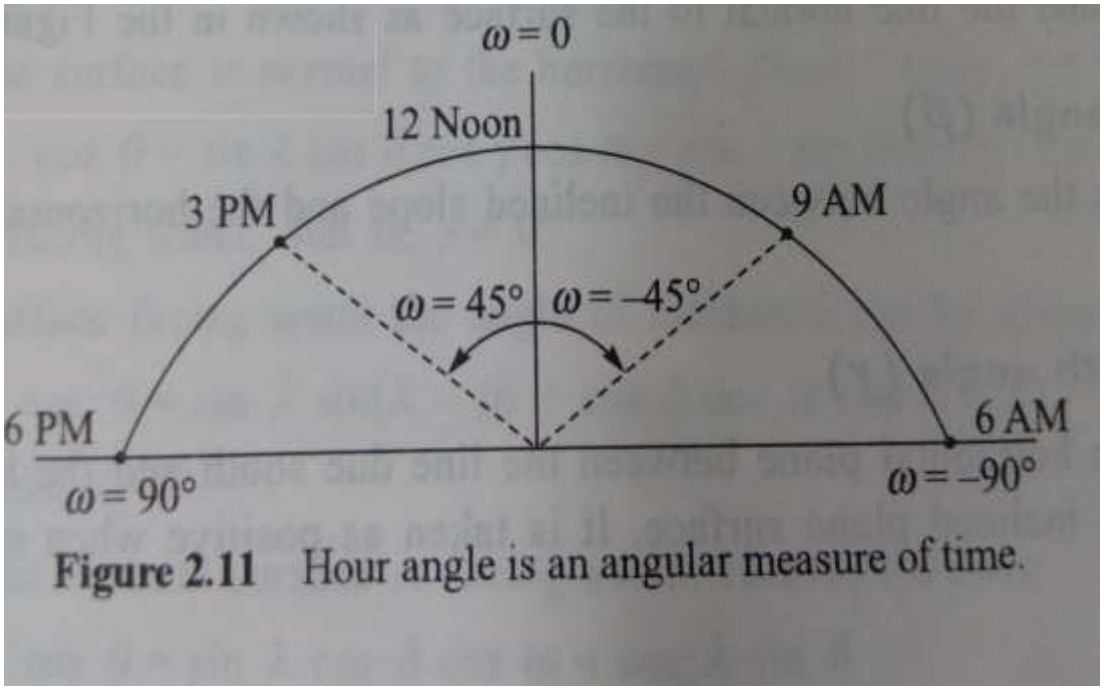


Figure 2.11 Hour angle is an angular measure of time.

- I hour corresponds to 15° of earth's rotation.(360° in 24 hour).
- As at solar noon the sun rays is in line with local meridian or longitude, the hour angle at that instant is zero.
- The hour angle can be given as:

$$\omega = [\text{solar time} - 12] \times 15^{o}$$

- Calculate the hour angle at 10:30 AM local apparent time.
- Solution:
- ω= [solar time -12]x 15°

=15*(10.5-12)

=-22.5^{o}

- **Calculate hour angle for the local apparent time given by (i) 6 AM, (ii) 9 AM, (iii) 12 noon, (iv) 3 PM, (v) 6 PM and (vi) 9 PM.**

The hour angle is given by

$$\omega = [\text{Solar time} - 12] \times 15^\circ$$

(i) Solar time = 6 AM

$$\omega = (6 - 12) \times 15^\circ$$
$$= -90^\circ$$

(ii) Solar time = 9 AM

$$\omega = (9 - 12) \times 15^\circ$$
$$= -45^\circ$$

(iii) Solar time = 12

$$\omega = (12 - 12) \times 15^\circ$$
$$= 0$$

(iv) Solar time = 3 PM = 15

$$\omega = (15 - 12) \times 15^\circ$$
$$= 45^\circ$$

(v) Solar time = 6 PM = 18

$$\omega = (18 - 12) \times 15^\circ$$
$$= 90^\circ$$

(vi) Solar time = 9 AM = 21

$$\omega = (21 - 12) \times 15^\circ$$
$$= 135^\circ$$

Inclination or altitude angle (α)

It is the angle between sun's ray and its projection on horizontal surface as shown in Figure 2.12.

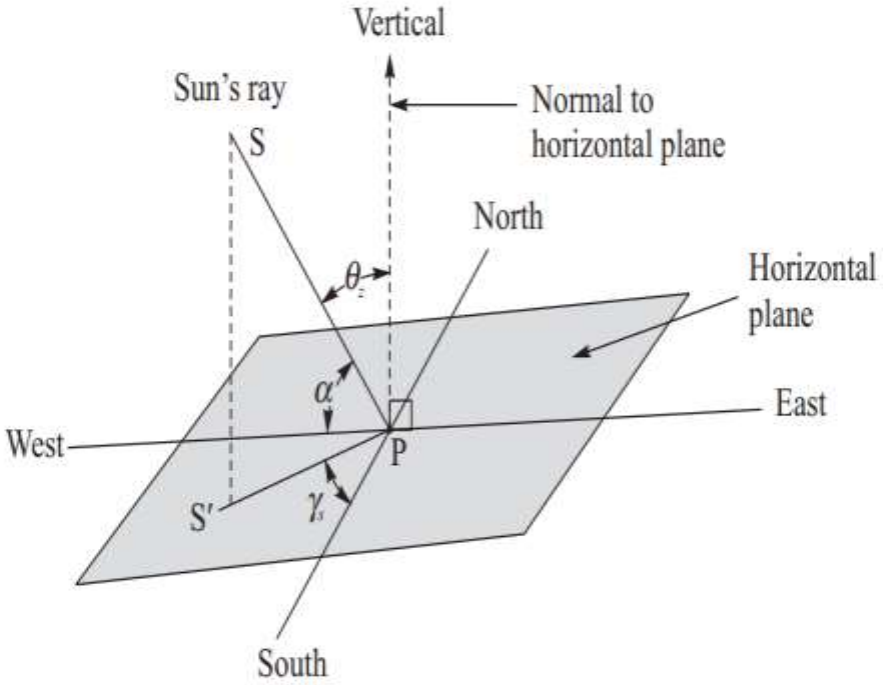


Figure 2.12 Inclination (altitude) angle, solar azimuth angle and zenith angle.

Zenith angle (θ_z)

It is the angle between sun's ray and normal to the horizontal plane as shown in Figure 2.12.

Inclination or altitude angle(αs)

- It is the angle between sun's ray and its projection on horizontal surface.

Angle of incidence(θ)

- The angle of incidence for any surface is defined as angle formed between direction of sun ray and line normal to surface.

Zenith angle(θz)

- It is the angle between sun's ray and normal to the horizontal plane.
- $\Theta z = \frac{\pi}{2} - \alpha$

Solar azimuth angle(γs)

- It is the angle between the projection of sun's ray to horizontal plane and line due south passing through that point. The value of the azimuth angle is taken positive when it is measured from south towards west.

$$\cos\theta_z = \sin\phi\sin\delta + \cos\phi\cos\delta\cos\omega$$

where Φ latitude angle

At sunrise and sunset θ=θz= 90 ^{o}

Cos θz = 0

Cos ω = -tanΦ tanδ

The angle between sunrise and sunset or solar day length= 2ω
The sun traverses 15 ^{o} in 1hour duration. The sunshine is available between sunrise and sunset in solar day length. The duration of sunshine hours or daylight hours can be given as

$$t_d = \frac{\text{Angle between sunrise and sunset}}{15^{o}} \times 1\,\text{h}$$

$$= \frac{2\times\omega}{15}\times 1$$

Angle of incidence(θ)

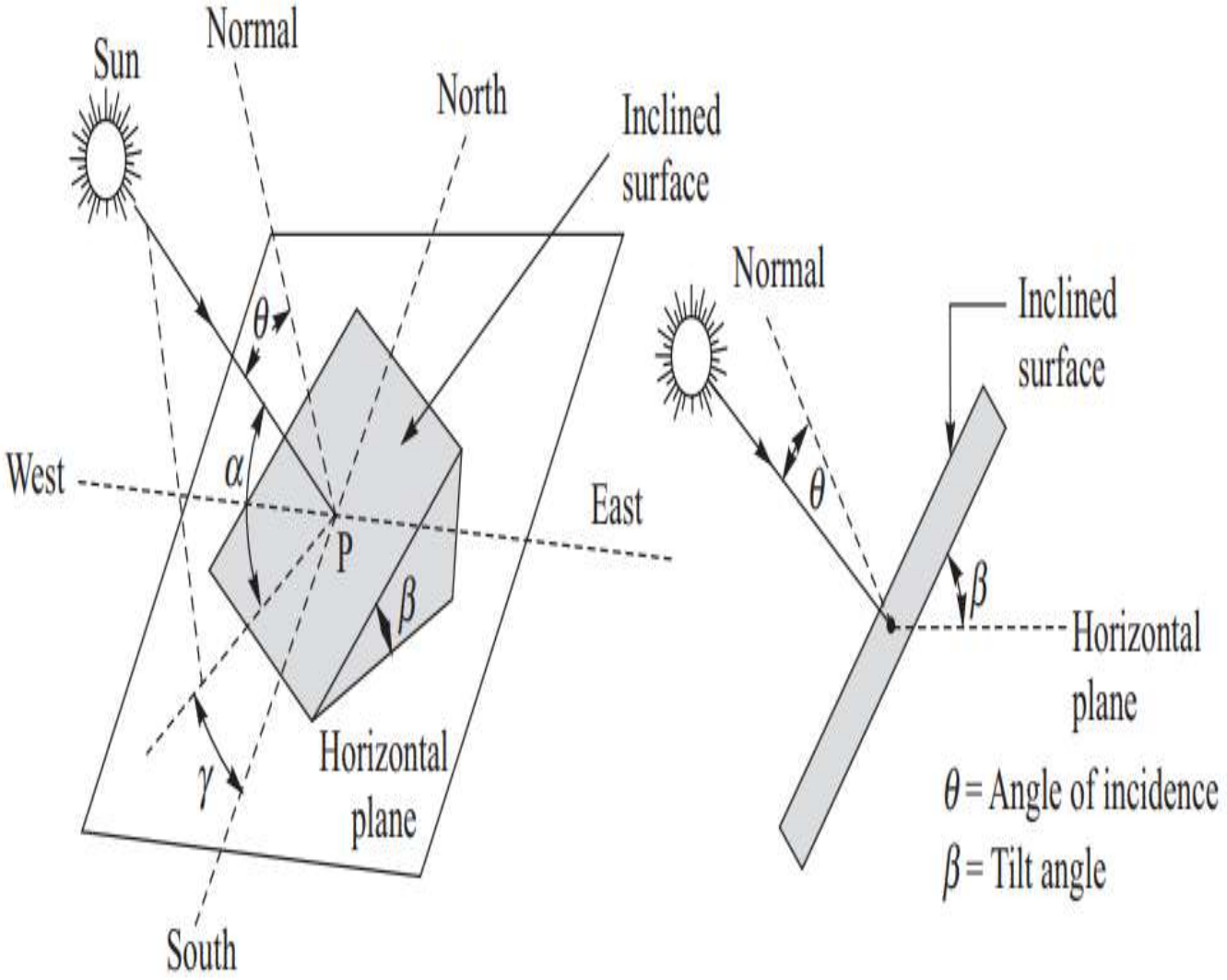
- The angle of incidence for any surface is defined as the angle formed between the direction of the sun ray and the line normal to the surface.

Tilt or slope angle(β)

- The tilt angle is the angle between the inclined slope and the horizontal plane.

Surface azimuth angle(γ)

- It is the angle in horizontal plane between the line due south and the horizontal projection of normal to inclined surfce.It is taken positive when it is measured from south towards west.


Sun
Normal
North
Inclined surface
θ
α
West
East
P
β
Horizontal plane
γ
South
Normal
Inclined surface
θ
β
Horizontal plane
θ = Angle of incidence
β = Tilt angle

- **Calculate the number of daylight hours in Srinagar for 1 January and 1 July. Take latitude of Srinagar as 34°05′ N.**

The declination angle δ is given as

$$\delta = 23.45 \times \sin\left[\frac{360}{365} \times (284 + n)\right]$$

where n is number of days of the year counted from 1 January.

Case I. 1 January

$$n = 1$$

$$\therefore \quad \delta = 23.45 \times \sin\left[\frac{360}{365} \times (284 + 1)\right]$$

$$= -23.01^\circ$$

λ latitude angle

$$\therefore \quad t_d = \frac{2}{15}\cos^{-1}[-\tan\lambda \times \tan\delta]$$

$$= \frac{2}{15}\cos^{-1}[-\tan(34.08) \times \tan(-23.01)]$$

$$= 9.77 \text{ h}$$

Case II. 1 July

$$n = 31 + 28 + 31 + 30 + 31 + 30 + 1$$

$$= 182$$

$$\delta = 23.45 \times \sin\left[\frac{360}{365}(284 + 182)\right]$$

$$= 23.12^{\circ}$$

$$t_d = \frac{2}{15}\cos^{-1}[-\tan(34.08 \times \tan(23.12)]$$

$$= 14.24 \text{ h}$$

- **Calculate the day length in hours in Mumbai on 1 December if latitude = 19.116 and declination angle = –22.108°.**

$$t_d = \frac{2}{15} \times \cos^{-1}(-\tan \lambda \times \tan \delta)$$

$$= \frac{2}{15} \times \cos^{-1}[-\tan 19.116 \times \tan (-22.108)]$$

$$= \frac{2}{15} \times 81.92$$

$$= 10.9 \text{ h}$$

- **Calculate hour angle at sunrise and sunset on 21 June and 21 December for a surface tilted at 10° and facing due south. The location has latitude 19° 07′ N and longitude 72° 51′ E.**

Case 1. 21 June

$$n = 31 + 28 + 31 + 30 + 31 + 21 = 172$$

$$\delta = 23.45 \sin\left[\frac{360}{365}(284 + 172)\right]$$

$$= 23.45°$$

$$\omega = \cos^{-1}\left[-\tan(\lambda - \beta) \times \tan\delta\right]$$

$$= \cos^{-1}\left[-\tan(19.12 - 10) \times \tan 23.45\right]$$

$$= 81.4$$

- **Calculate the zenith angle of the sun at a place (26.75° N) at 9.30 on 16 February.**

$$\omega = (9.30 - 12) \times 15 = -37.5°$$

$$n = 31 + 16 = 47$$

$$\delta = 23.45 \sin\left[\frac{360}{365}(284 + n)\right]$$

$$= 23.45 \sin\left[\frac{360}{365}(284 + 47)\right]$$

$$= -12.95$$

$$\cos\theta_z = \cos\delta \cos\lambda \cos\omega + \sin\delta \times \sin\lambda$$

$$= \cos(-12.95) \times \cos(26.75) \times \cos(-37.5) + \sin(-12.95) \times \sin(26.75)$$

$$= 0.589$$

$$\therefore \quad \theta_z = 53.92$$

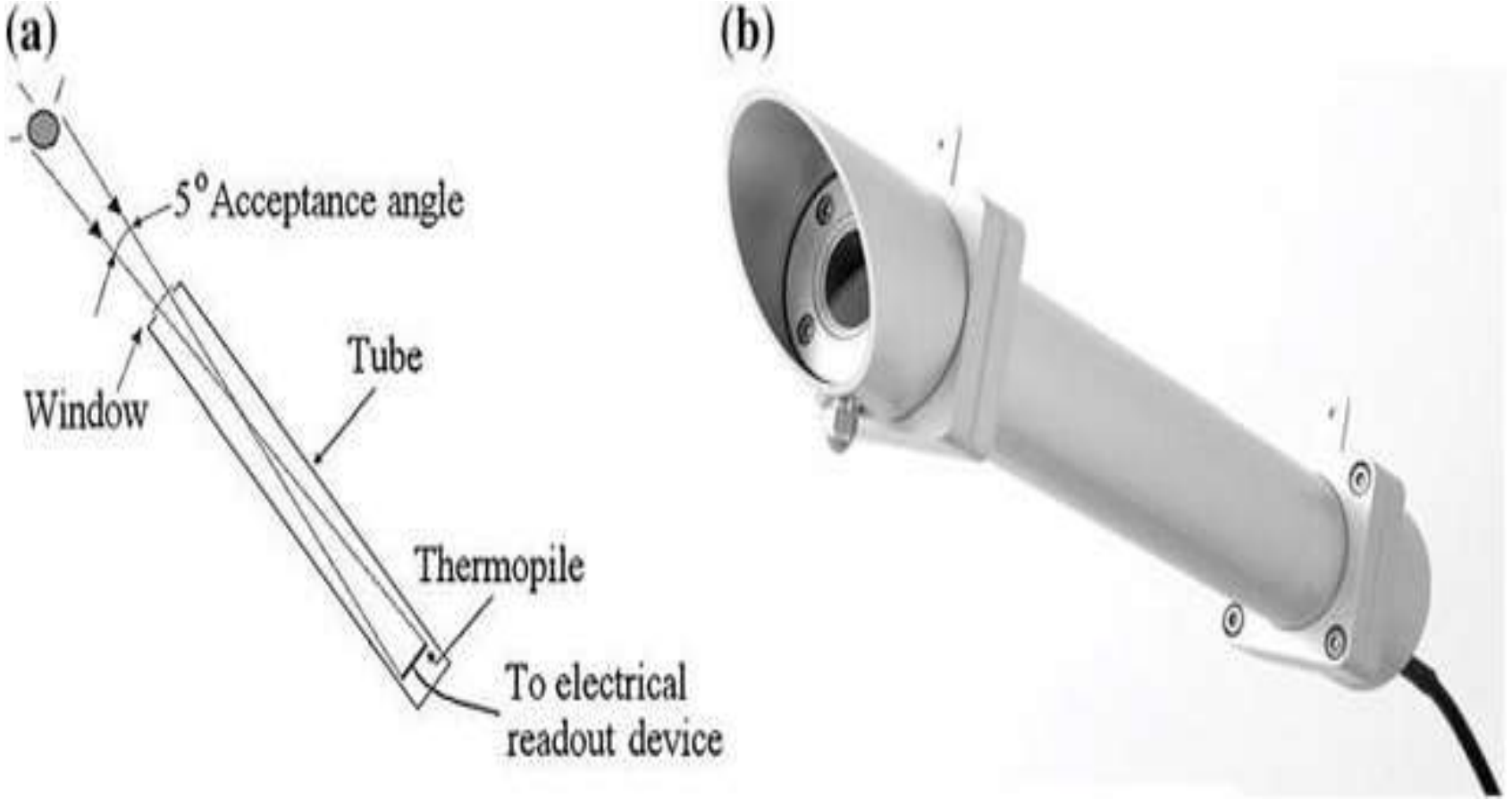
MEASUREMENT OF SOLAR RADIATION

Pyrheliometer-

is an instrument for measurement of direct beam solar irradiance. Sunlight enters the instrument through a window and is directed onto a thermopile which converts heat to an electrical signal that can be recorded. The signal voltage is converted via a formula to measure watts per square metre.

The sensor is a thermopile(A thermopile is basically a series arrangement of thermocouples used to develop a much greater voltage than its possible using only one.) and its disc is located at the base of a tube whose axis is aligned in the direction of sun's rays.

It is used with a solar tracking system to keep the instrument aimed at sun

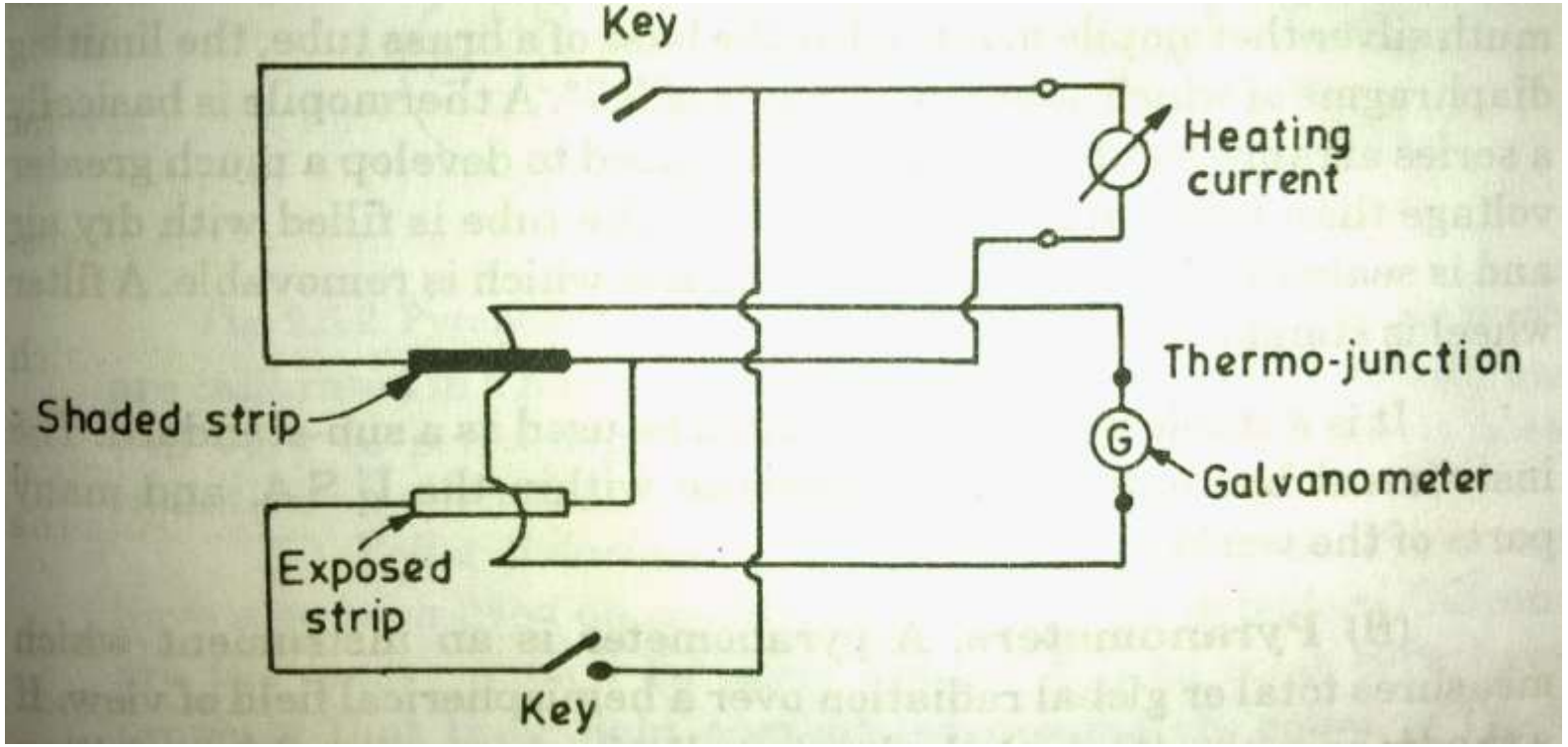

(a)
5° Acceptance angle
Window
Tube
Thermopile
To electrical
readout device
(b)

TYPES OF PYRHELIOMETERS

- Types :
  - Angstrom pyrheliometer
  - Silver disc pyrheliometer
  - Eppley pyrheliometer

Angstrom Compensation Pyrheliometer

- A thin blackened shaded manganin strip is heated electrically up to the same strip exposed to solar radiation.
- The thermocouples on the back of each strip connected in opposition through a galvanometer to check the equality of temperature.

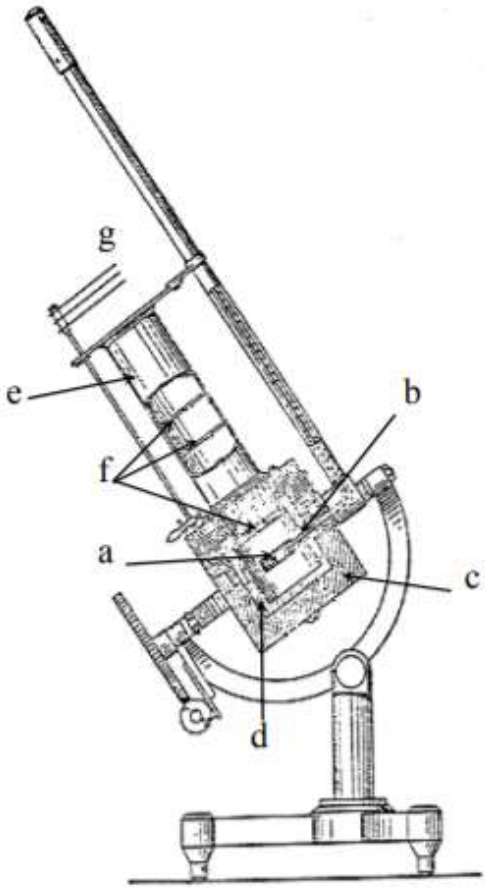


- When solar irradiance is measured with this type of pyrheliometer, the small shutter on the front face of the cylinder shields one sensor strip from sunlight, allowing it to reach only the other sensor.
- A temperature difference is therefore produced between the two sensor strips because one absorbs solar radiation and the other does not, and a thermo electromotive force proportional to this difference induces current flow through the galvanometer.
- Then, a current is supplied to the cooler sensor strip (the one shaded from solar radiation) until the pointer in the galvanometer indicates zero, at which point the temperature raised by solar radiation is compensated by Joule heat.

- Energy H of direct radiation $H_{DN}=Ki^2$
- H_{DN}= Direct radiation incident on an area normal to sun's rays, i= heating current
- K is a dimension and instrument constant
- $=\frac{R}{W\alpha}$ R is the resistance per unit length of the absorbing strip, w mean width of absorbing strip , α absorbing coefficient of absorbing strip

Silver Disk Pyrheliometer

- The sensing element is a silver disk measuring 28 mm in diameter with a thickness of 7 mm that is painted black on its radiation-receiving side.
- It has a hole from the periphery toward the center to allow insertion of the bulb of a high-precision mercury-in-glass thermometer.
- To maintain good thermal contact between the disk and the bulb, the hole is filled with a small amount of mercury. It is enclosed outside by a heat-insulating wooden container.
- The stem of the thermometer is bent in a right angle outside the wooden container and supported in a metallic protective tube.
- A cylinder with diaphragms inside is fitted in the wooden container to let direct solar radiation fall onto the silver disk.
- There is a metallic-plate shutter at the top end of the cylinder to block or allow the passage of solar radiation to the disk.



a: Black-painted silver disk
b: Mercury-in-glass thermometer
c: Wooden container
d: Copper case
e: Cylinder
f: Diaphragm
g: Shutter

- During the measurement phase, the disk is heated by solar radiation and its temperature rises. The intensity of this radiation is ascertained by measuring the temperature change of the disk between the measurement phase and the shading phase with the mercury-in-glass thermometer.

Eppley Pyrheliometer

- Bismuth-silver thermopile with 15 temperature compensated junctions mounted at the base of a brass tube with a series of diaphragms.
- The tube is filled with dry air and sealed with crystal quartz window which is removable.
- The instrument is mounted on a motor-driven heliostat which is adjusted every week to cover changes in the sun’s declination



Pyrheliometer

Advantages

The **advantages of the Pyrheliometer** include the following.

- Very low power consumption
- Operates from a wide range of voltage supplies
- Ruggedness
- Stability

Pyrheliometer Applications

The applications of this instrument include the following.

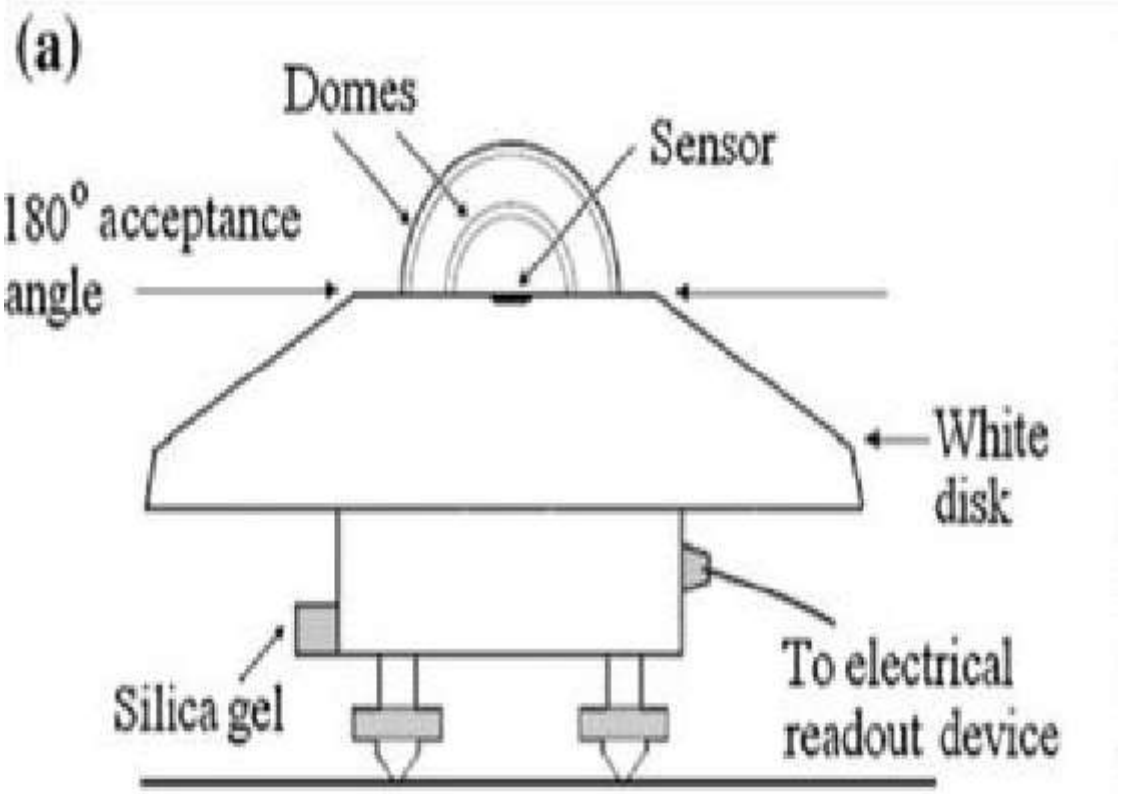
- Observations of Climate
- Testing research of Material
- Estimation of the solar collector’s efficiency
- PV devices

Pyranometer

- A **pyranometer** is used for measuring solar irradiance on a planar surface



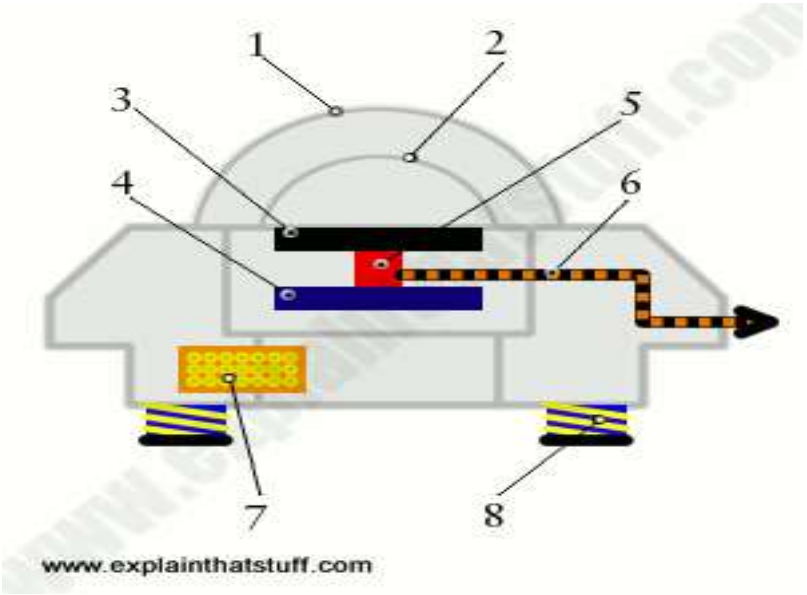
Pyranometer



- The Acceptance angle for solar irradiance measurement is the maximum angle at which incoming sunlight can be accurately measured or captured by an instrument or solar concentrator.

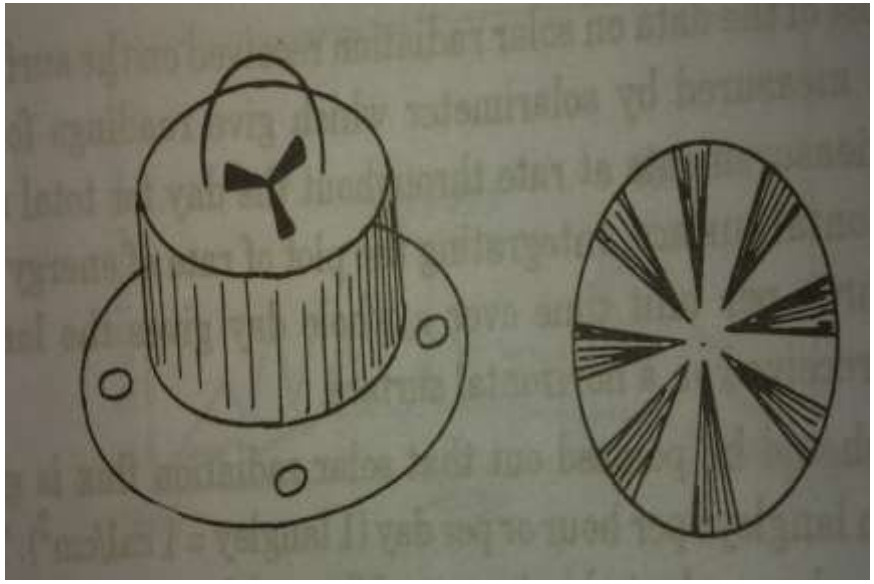
- The 180-degree field of view allows the instrument to capture incoming radiation from all directions above the plane of the sensor, from horizon to horizon.

1. Outer dome made from a hemisphere of optical-quality glass.
2. Inner dome made from a smaller hemisphere of optical glass.
3. Black carbon disk (illuminated by the Sun) absorbs a broad range of wavelengths of solar radiation and acts as the sensing element.
4. Second, control disk (not illuminated by the Sun) acts as a comparison and compensating element.
5. Thermopile temperature sensor compares the temperature rise of the two disks.
6. Output lead (usually about 10m or 30ft long).
7. Replaceable silica gel cartridge (dessicant) absorbs moisture to prevent dew forming inside on cold nights.



Eppley Pyranometers

- It uses concentric silver rings 0.25 mm thick, appropriate coated black and white with either 10 or 50 thermo couple junctions to detect temperature differences b/w coated rings
- There is a temperature difference between black surface (absorbs most solar radiation) and white surface (reflects most solar radiation). Detection of temperature difference achieved by thermopile





Yellot Solarimeter (Photovoltaic solar cell)

- used on photovoltaic(solar cell) detectors.
- Silicon cells are used to measure solar radiations.
- The incident radiations on solar cells are converted to the equivalent electrical energy.

Advantages

- The temperature coefficient is extremely small
- Standardized to ISO standards
- Measurements are accurate

Disadvantages

- Response time is longer compared to PV cell.
- The disadvantage of the pyranometer is, its spectral sensitivity is imperfect, so it does not observe the complete spectrum of the sun. So errors in measurements can occur.

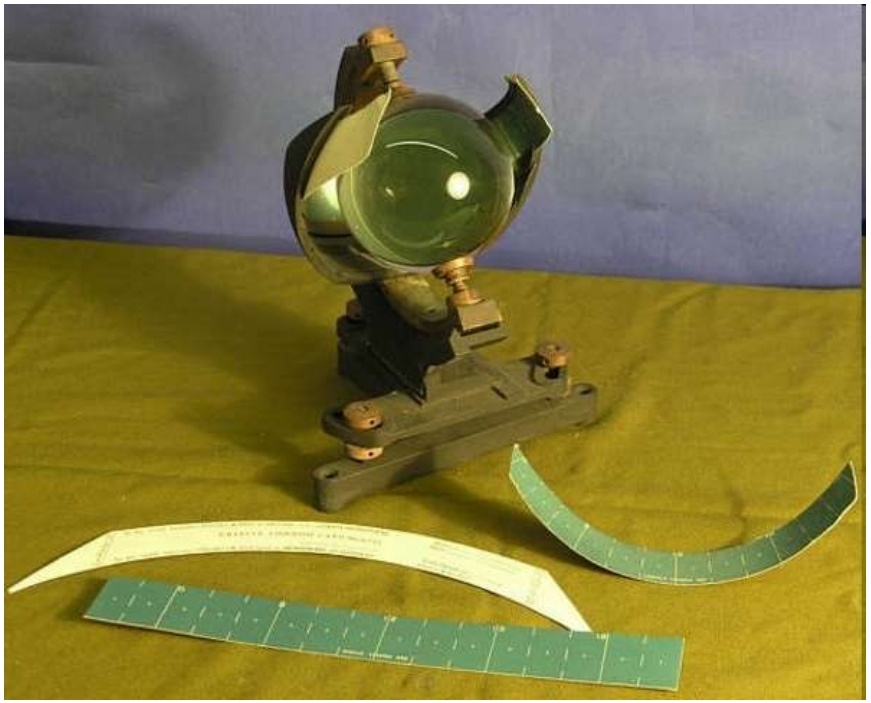
Pyranometer Applications

The applications are

- The solar intensity data can be measured.
- Climatological & Meteorological studies
- PV systems design
- Locations of the greenhouse can be established.
- Expecting the requirements of insulation for building structures

Sunshine Recorder

The duration in hours of bright sunshine in a day is measured by a sunshine recorder



- It consists of a glass sphere installed in a section of spherical metallic bowl, having grooves for holding a recorder card strip.
- The glass sphere is adjusted to focus sun rays to a point on the card strip.
- On a bright sunshine day, the focused image burns a trace on the card.
- Through the day the sun moves across sky, the image moves along the strip.
- The length of image is a direct measure of the duration of bright sunshine.

- Components: A glass sphere (lens) mounted in a metal bowl with slots for different cards, and a base adjustable for latitude.
- Mechanism: The glass sphere acts as a lens, focusing the sun's rays onto a specific point on the card.
- Recording: As the sun moves across the sky, the focused spot burns a track on the card.
- Card Types: Different cards (straight, short curved, long curved) are used for different seasons, and the total hours are read from the burn length.